Group 4 - Team Statement

Milestone 1: Twenty-table core and work plan

Updated October 15, 2026. Four members, five tables each. English names, acknowledgment, repository access and meeting details remain pending.

Shared understanding

CloudNine connects client requests to physical resources, deployments, usage and charges. Our demonstration uses synthetic activity and two read-only website pages: Operations and Billing. It does not provision real servers. Each member must be able to explain the complete reservation-to-invoice example.

The model keeps 20 tables. Independent projects and many-to-many client membership are outside scope. charge_share doubles as an invoice line, with the payer derived from invoice. Data centers remain because location filters, capacity comparisons and allocation constraints are part of the scenario and planned website.

Member A - Clients and reservations

Own client, user_account, reservation, reservation_item and billing_account. Implement client hierarchy, a user's optional current client, requests and payer contacts. Check item quantities, location restrictions and client ownership. Prepare synthetic clients, staff, reservations and billing accounts. Own Q3 reservation shortages and the Operations reservation/detail section. Draft scenario, client assumptions and the combined report. Review Member D's payer and invoice joins.

Member B - Infrastructure

Own region, availability_zone, data_center, resource_type and resource. Implement the location hierarchy, scoped resource identity and scalar capacities. Prepare multiple locations, resource types and realistic capacities. Own Q1 capacity by location and Q2 utilization by resource type; build Operations location filters and capacity summaries. Integrate the ERD source and shared schema. Review Member A's requested types and location joins.

Member C - Deployments and usage

Own deployment, resource_allocation, resource_state, deployment_service and usage_record. Implement interval records, replacement lineage, service provisioning and estimated/actual consumption. Prepare maintenance, shortages, substitutions and consumption spanning the sample quarter. Own Q4 downtime and Q5 substitutions; build the Operations deployment and availability sections. Co-own Q6 usage comparison with D. Generate and inspect final PDFs/SVGs. Review Member B's capacity and resource identity.

Member D - Services and billing

Own service, service_rate, charge, charge_share and invoice. Implement rate history, charge contexts, credits, draft invoice shares and immutable issued totals. Prepare rates, fees, credits, split payers and invoices. Own Q7 client charge trends and Q8 payer invoices; build the Billing page. Co-own Q6 with C, who supplies consumption records. Draft billing/normalization reasoning and maintain the submission checklist. Review Member C's metering and deployment joins.

Balanced work and integration

Five tables each is a starting allocation, not a guarantee of equal effort. C's temporal logic and D's billing rules are comparatively demanding; A assembles the report and B integrates the schema and reusable website layout to balance that work. Each owner writes their module's data, queries, validation and explanation. C coordinates document generation but all four inspect the outputs.

Use one shared database and common date/location filters. Agree on the DBMS and repository structure before implementation. Each owner supplies data that references the others' fixtures correctly. Hold a joint review of the reservation-to-invoice example before connecting pages. Query numbers refer to Group4_Website_Plan.md and must remain consistent across source, report and presentation.

At each check-in, compare estimated remaining effort. If a member has substantially more work, transfer a specific query, data-generation task or page section with an explicit owner and reviewer. A and B can take fixture validation or page integration from C and D. Keep a short contribution log; commit counts alone do not measure effort.

Review and accountability

Create a GitHub Issue for each concrete task with its owner, reviewer, expected output and deadline. A reviews D, B reviews A, C reviews B, and D reviews C. Use branches and pull requests for later group implementation. A task is complete only after its artifact works, feedback is addressed and the assigned reviewer confirms it.

Hold two short weekly check-ins at times chosen by the members. Report blockers promptly and pair with the reviewer to resolve them. Record assumptions and workload changes in the repository. Meetings, names and member acceptance are not represented as already completed.

Diagram and website standards

Use singular lower_case tables and camelCase attributes. Display only PK/FK on the ERD; keep uniqueness and cross-table rules in analysis R1-R9. Every relationship has a name and cardinalities. GitHub previews use the same four SVGs as the PDF. Each website metric must have a documented query, units, selected time period and interpretation. Avoid summing estimates with actuals, double counting repeated joins, or mixing unrelated billing units.

Milestones and submission

Milestone 1 includes the four-page ERD, analysis report and this team plan. Later work implements all 20 tables, realistic synthetic records across at least a quarter, eight distinct queries and two read-only pages. The website plan is preparation, not a claim that queries or pages are already implemented.

Before submission, all members inspect documents, verify links and repository access, and confirm their assigned work. Map A-D to actual English names and GitHub accounts. Communication channel, course deadline, meeting schedule and internal review date remain to be supplied. Aim for a combined review at least 48 hours before the course deadline when feasible.